THE SURROGATE GAP: WHEN DO LABEL-FREE BEHAVIORAL TESTS TRACK HUMAN AGREEMENT FOR LLM JUDGES?

Anonymous authors
Paper under double-blind review

ABSTRACT

When a language-model judge scores AI-generated Lean 4 theorems against gold references, how much should we trust the judge itself? Re-validating every model and prompt with fresh human ratings does not scale, so we test whether validation can be *amortized*: collect a fixed human criterion once, then evaluate new judges with falsifiable, label-free behavioral probes. We separate scoped trustworthiness TW_m (construct), human agreement TI_{m_1} (criterion), and the exact frozen proxy $\mathrm{PTI}_{\mathrm{v0}}$ (instrument). $\mathrm{PTI}_{\mathrm{v0}}$ is the zero-first geometric mean of equality, bug-monotonicity, missing-specification monotonicity, and alpha-invariance properties; passing is a screen result, not proof of trustworthiness. Our complete subscription-CLI experiment evaluates five primary judge–prompt configurations and three controls with three repeats per score cell (774 CLI tasks; 4,728 cells; no missing scores). Four of five primary $\mathrm{PTI}_{\mathrm{v0}}$ point estimates are exactly zero; the remaining configuration scores 0.811. The pre-registered outer bridge is not positive: primaries-only Spearman is -0.354 (95% judge-bootstrap interval $[-1.000, 0.395]$, n=5), below the 0.5 smallest effect size of interest and inconclusive about zero. A frozen simulation gives only 27.1% power for the complete positive decision rule at true $\rho = 0.5$. Historical development-panel $\mathrm{PTI}_{\mathrm{var}}$ associations are retained only as LEGACY—NOT $\mathrm{PTI}_{\mathrm{v0}}$; criterion drift ($\rho \approx 0.13$ across annotation protocols) and a registered 32-judge negative further bound transfer. The result is an audited failure to support the registered positive bridge for this frozen proxy panel and a protocol for stronger tests, not evidence that human validation can yet be replaced.

1 INTRODUCTION

Agentic AI systems increasingly generate both code and the trust artifacts meant to evaluate that code: tests, specifications, and formal theorems. In Lean 4, these artifacts are code-adjacent formal objects: runnable programs paired with theorem or specification statements and, sometimes, proofs. In classical autoformalization the target theorem is supplied, so evaluation can reduce to proof success; in agentic verification the system must decide *what should be true* before proving anything. The Lean kernel can check a proof, but it cannot say whether a proposed theorem is complete, semantically aligned with the intended program, or merely a weak statement that typechecks. As agents produce more candidate artifacts than humans can inspect, theorem-quality evaluation becomes a code-evaluation and scalable-oversight problem (Bowman et al., 2022; Zhou et al., 2025; Khattab et al., 2023; Cheng et al., 2024).

The standard reliability check is per-judge human validation: collect labels, measure agreement between judge and human ratings, and keep the judge if agreement is high. That check is necessary but scales linearly with the number of judges, prompts, model updates, thresholds, and domains; every new model family or prompt revision asks for another round of annotation. Humans remain the anchor — the waste is re-using them from scratch for every judge. Our question, in one sentence: *can automatically measured behavioral properties predict a scoped human-alignment criterion for unseen LLM judges, which would amortize — not eliminate — human annotation if a bridge succeeds?*

Following the validity-centered evaluation framework of Salaudeen et al. (2025), we separate the *construct* (scoped judge trustworthiness, TW_m), the *criterion* that measures it on a specified panel (a human-grounded index TI_m), and the *instrument* (a label-free property vector $\mathbf{P}$ with aggregate PTI_p). The properties are concrete: a judge should give maximal scores to identical artifacts, lower scores as semantic bugs accumulate, lower scores as specification elements are removed, and stable scores under repeats and semantics-preserving rewrites. The instrument has two jobs with different success metrics (Section 5): a *screen* that gates judges before humans are spent, and a *surrogate* that predicts the criterion numerically. Passing the screen never establishes trustworthiness by itself; acceptance is licensed only by a demonstrated bridge between PTI_p and TI_{m_1}, and only within the validated judge population.

Our **contributions** are:

1. We distinguish scoped judge trustworthiness TW_m (construct), its human-grounded criterion TI_m measured on a specified panel, and the automated behavioral instrument $\mathbf{P}(aj)$, making explicit which claims need human labels and which do not (Section 3).

2. We introduce a reusable suite of falsifiable behavioral probes for LLM judges of formal-code artifacts, with scripted perturbation ladders and pre-specified estimators (Section 4).

3. We freeze and execute $\mathrm{PTI}_{\mathrm{v0}}$ on five primary judge–prompt configurations and three controls. The pre-specified primaries-only validation bridge is Spearman -0.354 (95% interval $[-1.000, 0.395]$, $n{=}5$), below the 0.5 SESOI and inconclusive about zero; no positive surrogate or deployment claim is supported (Section 6.3).

4. We disclose the complete score/probe audit, low-power simulation, model-execution limitations, criterion drift ($\rho \approx 0.13$ between labeling campaigns), and the separate pre-registered 32-configuration negative, deriving the protocol requirements they impose (Section 7).

2 Related Work

LLM-as-a-judge. LLMs are increasingly used as automated evaluators, with reliability commonly validated by agreement with people: G-Eval reports Spearman correlation with human scores (Liu et al., 2023), MT-Bench reports judge agreement with human preferences (Zheng et al., 2023), and Judge-Bench evaluates replication of human annotations across 20 NLP datasets (Bavaresco et al., 2024). Large-scale audits caution that reliability varies by task and setup and stress explicit validation before replacing human judgment; known biases include position, verbosity, and prompt sensitivity (Bavaresco et al., 2024; Li et al., 2024; Zheng et al., 2023). We keep human agreement as the criterion and test a possible amortization: one development panel relates falsifiable behavioral properties to human agreement, after which new configurations could be screened without new labels only if the bridge succeeds. The frozen result below does not license that step. Closest to this paper, a non-archival workshop study certifies a Lean-aware theorem judge with identity and monotonicity checks (Anonymous, 2025); this paper recasts that protocol in construct/criterion/instrument terms and adds screen-versus-surrogate semantics, audited data disclosures, and pre-registered held-out tests.

Improving and evaluating judge reliability. Debate-style panels (Chan et al., 2024), rubric-driven prompting (Liu et al., 2023), and internal-consistency measurement (Li et al., 2024) target reliability at inference time. For code, ICE-Score and CodeJudge highlight the difficulty of one-shot correctness assessment (Zhuo, 2024; Tong & Zhang, 2024). Consistency-focused analyses argue stability, not reasoning complexity, drives reliable grading (Chen et al., 2026). Our probes operationalize exactly this: identity, monotonicity under controlled edits, and stability under rewordings, in the spirit of metamorphic testing. Salaudeen et al. (2025) provide the construct/criterion/instrument vocabulary we adopt; Robertson & Koyejo (2025) give information-theoretic oversight queries for the verifier-absent regime, complementary to our verifier-present setting. Appendix J surveys scalable oversight, consensus aggregation (Denisov-Blanch et al., 2026), test-set-specialized autoraters (Finkelstein et al., 2025), autoformalization, and HCI perspectives.

3 Problem Setup and Definitions

Task. Each benchmark item contains a natural-language problem statement, a gold Lean 4 artifact g, and a candidate artifact x produced by an AI agent; artifacts may include code, tests or specifications, and a target theorem/specification statement. An LLM judge parameterized by $aj = (\text{model}, \text{prompt}, n_{\text{repeats}}, \text{aggregation})$ maps (x, g) to a score $\hat{s}_{aj}(x, g) \in [0, 1]$ reflecting how well the candidate captures the intended formal specification. Type-checking alone cannot decide this: a candidate may compile while being under-specified, over-constrained, vacuously true, or semantically shifted (Miranda et al., 2026), and no complete mechanical check of specification equivalence exists (Rice, 1953).

Three objects, three roles. Earlier versions of this work collapsed the following into one scalar; we keep them separate throughout (role labels follow Salaudeen et al., 2025):

Object	Role	Notation	Needs labels
Trustworthiness (predicate we care about)	construct	$\text{TW}_m(aj)$	—
Trust index (criterion on a specified panel)	criterion	$\text{TI}_m(aj)$	yes
Proxy trust index (behavioral aggregate)	instrument / surrogate	$\text{PTI}_p(aj)$	no

Table 1: **Construct, criterion, instrument.** Subscripts follow the L_p convention: the capital letter names the family, the subscript selects the variant. m indexes the matching criterion on the target side; $p \subseteq \{1, \ldots, K\}$ indexes the property set on the proxy side and is locked before any bridge analysis (choosing p to maximize the bridge correlation is model selection over $2^K - 1$ subsets and reintroduces selection circularity).

The m-ladder. m_0: distributional match, $P_\theta(s \mid \cdot, aj) \approx P_H(s \mid \cdot, h_j)$. m_1: correlational — $\text{Corr}(S_{aj}, S_{h_j})$ high over items. m_2: m_1 plus stability — $\text{std}(\hat{s}_{aj})$ low. We write $\text{TW}_2 \equiv \text{TW}_{m_2}$; m_2 is the working target. The construct is

$$\begin{aligned}\text{TW}_m(aj) :\ &\forall a \in L_A,\ \text{if } aj \approx_\Pi h_j \text{ then} \\ &\big\| P_\theta(s \mid x, y^*, \hat{y}{=}a(x), aj) \approx_m P_H(s \mid x, y^*, \hat{y}{=}a(x), h_j) \big\|_m \text{ small},\end{aligned} \tag{1}$$

where $aj \approx_\Pi h_j$ is the *persona precondition*: the judge is conditioned toward the rater persona $h_j = (\text{person}, \pi)$, without which the definition is unsatisfiable over a heterogeneous rater pool. Items are what the m_1/m_2 criteria integrate over — they are never universally quantified in a correlational statement. The measurable objects are

$$\text{TI}_{m_1}(aj) = \text{Corr}_{i \in \text{panel}}\big(\hat{s}_{aj}(i),\ s^*_{h_j}(i)\big), \qquad \text{PTI}_p(aj) = \text{GM}\big(P_k(aj) : k \in p\big), \tag{2}$$

and at the m_2 target the criterion gains a stability conjunct,

$$\text{TI}_{2'}(aj) = \Big(\text{clip}_{[0,1]}\big(\text{Corr}(D_{\hat{S}_{aj}}, D_{\hat{S}^*_H})\big) \cdot \big(1 - \text{STD}[D_{\hat{S}_{aj}}]\big)\Big)^{1/2}, \tag{3}$$

with the stability term judge-side (it is P_4, the label-free conjunct), orientation-flipped, and the correlation clipped before the geometric mean. Correlations are computed only over *paired* observations — the same candidate scored by both sides — with repeats aggregated first: the K_a agent generations, K_{aj} judge repeats, and K_h human passes are three distinct indices that never align pairwise. The exact headline instrument in this paper is PTI_{v0}, whose four components and numerical rules were frozen before the new judge scores and before the bridge statistic (Section 4). The item-level $\text{TI}_{2'}$ diagnostic remains a decomposition of criterion validity and judge repeat stability; it is not the target axis of the outer bridge. The older $\text{PTI}_{\{1,2,3\}}$ and variance-aware PTI_{var} aggregates survive only as historical development-panel analyses and are never relabeled PTI_{v0}.

Three claims. *Claim A* (definitional): aj is TW_2-trustworthy — needs humans. *Claim B* (proxy): $\text{PTI}_p(aj) \geq \tau$ — label-free. *Claim C* (bridge): $\text{PTI}_p \approx \text{TI}_{m_1}$ over held-out judges — established once, population-scoped. Failing B is evidence against A with strength set by C; passing B never establishes A except through C. Refuting A from a single probe violation requires one of two repair routes. *Route A (definitional)*: define $\text{TW}(aj) := [\text{TI}_{m_1} \geq \tau_A] \wedge \bigwedge_k [P_k \geq \tau_k]$, so a probe violation rejects by definition and "properties predict alignment" is reported as a separate empirical result. *Route B (conditional falsification)*: under stated metamorphic-validity and margin assumptions, sufficiently large probe violations falsify pointwise ε-alignment (Lemma 1, Appendix C); the

margins δ are currently assumed, not measured, and a small human collection on perturbed ladder items would ground the lemma while also testing whether ALPHA-renaming is human-invariant — the standing P_4 failure mode. We claim no "sound rejection by construction" at the ratified target: under TW_2, correlation is an aggregate, so a judge can carry high TI_{m_1} while exhibiting a local bug-ladder reversal, and P_k low does not logically imply TI_{m_1} low.

Amortization and its four generalization axes. The criterion is only measurable on the annotated subset $b' \subseteq b$; the proxy runs on the observed typed pool: $\mathrm{PTI}(b, L_A, aj, P) \approx \mathrm{TI}_{2'}(b', L_A, aj, H)$. This exposes four axes that any amortization claim must name: held-out *judges* (the primary target, but not a generalization claim of the current validation analysis), new *agents*, new *benchmarks*, and within-benchmark *item-level extrapolation* $b' \to b$ — here the bridge is evaluated on 72 items while the proxy is evaluated on 470 observed typed candidates over 157 items (a nominal 157×3 grid with one missing candidate). Conditional on a future successfully established bridge, the cost decomposition is

$$C_{\text{amortized}} = \underbrace{|\Pi|\, n\, c_{\text{human}}}_{\text{persona conditioning}} + \underbrace{|AJ_{\text{val}}|\, |I|\, c_{\text{human}}}_{\text{bridge, once}} + \underbrace{|AJ|\, |A|\, |Q|\, c_{\text{API}}}_{\text{per screened judge}}, \tag{4}$$

Only the human-annotation terms are $O(1)$ in subsequently screened judges for fixed personas and a fixed validation panel; the inference term remains linear in $|AJ|$. The present bridge does not support this route and demonstrates no savings, so "no new human labels per screened judge" and "human-label amortized" are future-conditional claims, not results of this study.

4 Behavioral Probes, Estimators, and Data Pipeline

The four properties. *P1 — identity & consistency*: the judge should assign the maximum score to an artifact compared with itself, $s(X, X) = 1$. *P2 — monotonicity to bugs*: scores should decrease as semantic bugs are cumulatively introduced. *P3 — monotonicity to missing specifications*: scores should decrease as theorems/tests are removed. *P4 — stability & semantic invariance*: scores should remain stable across repeated calls and under semantics-preserving rewrites; ALPHA-renaming changes only bound-variable names, so a robust judge should score original and renamed artifacts almost identically. We evaluate these with rank-based monotonicity statistics (Kendall's τ, Spearman's ρ) and pairwise ordering accuracy, following prior judge-evaluation work (Tong & Zhang, 2024; Zhuo, 2024).

Frozen $\mathrm{PTI}_{\mathrm{v0}}$ indices and local aggregation. A typed base artifact is $f = (i, a, q)$: benchmark item i, generating agent a, and generation index q. Judge repeats use r, human pass/rater observations use u, and ladder degradation uses ℓ; these axes are never pooled positionally. For property type T, $\widetilde{s}_{f,T,\ell} = \mathrm{median}_r\ s_{f,T,\ell,r}$, with larger ℓ meaning greater degradation. For each structurally valid bug or missing-specification ladder,

$$\rho_{f,T} = \mathrm{Spearman}_\ell(\ell, \widetilde{s}_{f,T,\ell}), \quad C_{f,T} = \frac{1 - \rho_{f,T}}{2}, \quad \sigma_{f,T} = \mathrm{mean}_\ell(2\ \mathrm{sd}_{r,\mathrm{ddof}=0}\, s_{f,T,\ell,r})\,, \tag{5}$$

and $\mu_{f,T} = \sqrt{C_{f,T}(1 - \sigma_{f,T})}$. If $K_{aj} = 1$, stability is unobserved and $\mu_{f,T} = C_{f,T}$. If all ladder medians are exactly constant, the ladder remains a behavioral observation with $C_{f,T} = 0$; it is not dropped as an undefined correlation. Spearman uses average ranks for ties and endpoint snapping within 10^{-12}; Kendall τ_b, ties, reversals, and constant flags are diagnostics.

Four properties and exact-zero aggregation. The frozen components are

$$P_{\mathrm{eq}} = \mathrm{GM}_f(\mathrm{median}_r\, s(x_f, x_f))\,, \tag{6}$$

$$P_{\mathrm{bm}} = \mathrm{GM}_f(\mu_{f,\mathrm{bug}}), \qquad P_{\mathrm{ms}} = \mathrm{GM}_f(\mu_{f,\mathrm{missing\ spec}}), \tag{7}$$

$$P_{\mathrm{eq}\alpha} = \mathrm{GM}_f(1 - |\mathrm{median}_r\, s(x_f, g_i) - \mathrm{median}_r\, s(\alpha(x_f), g_i)|)\,, \tag{8}$$

$$\mathrm{PTI}_{\mathrm{v0}} = \mathrm{GM}(P_{\mathrm{eq}}, P_{\mathrm{bm}}, P_{\mathrm{ms}}, P_{\mathrm{eq}\alpha}). \tag{9}$$

Here α renames binders and all bound occurrences consistently and is the only top-level invariance transform. Every geometric mean is computed in the log domain after an exact-zero check: any observed zero factor makes the aggregate exactly zero, with no epsilon floor. Missing, non-finite, or unparsable scores are never zeros.

Probe validity is frozen independently of judge output: missing, duplicate/no-op, malformed, or non-elaborating transformations are excluded globally. After that structural freeze, required score coverage is 100%; insufficient coverage makes the property `NOT_COMPUTABLE` and the screen `DEFER`. Property and $\mathrm{PTI}_{\mathrm{v0}}$ intervals use 10,000 percentile bootstrap replicates clustered by f with shared resampling across properties. The criterion is $\mathrm{TI}_{m_1}(aj) = \mathrm{Spearman}_{i\in 72}(\mathrm{median}_r\, \hat{s}_{aj,i}, \mathrm{mean}_u\, s^*_{H,i,u})$, with strict and lenient views secondary and 10,000 paired item-cluster bootstrap replicates. The outer bridge is judge-indexed, $\{(\mathrm{PTI}_{\mathrm{v0}}(aj_j), \mathrm{TI}_{m_1}(aj_j))\}_{j=1}^{n}$: primaries-only Spearman is primary, Kendall τ_b is robustness, controls are secondary, and undefined judge-bootstrap replicates are dropped and counted. The axes are not a common measurement scale, so we use neither an identity line nor level MAE.

The SCREEN workflow. (1) Label a fixed panel once; (2) evaluate P1–P4 for bridge-set judges on scripted perturbations; (3) establish the bridge with uncertainty; (4) screen new configurations by property checks alone — FAIL-SCREEN on a per-property violation, PASS-SCREEN licenses acceptance only through the bridge within the validated population, DEFER otherwise. The present result stops at step (3) with DEFER; it does not license step (4).

5 TWO JOBS, TWO VALIDATIONS

The instrument does two jobs that require different evidence, and the paper never conflates their metrics.

	Screen (gate before humans)	**Surrogate (predict the criterion)**
Claim	catches unacceptable judges cheaply	PTI_p tracks TI_{m_1} numerically
Decision	FAIL-SCREEN / PASS-SCREEN / DEFER	predicted score with interval
Metrics	false-pass rate, false-reject rate, defer rate, coverage, human ratings saved at a fixed false-pass tolerance	held-out rank correlation, MAE/score-scale agreement, prediction intervals, transfer across families/prompts/personas/agents

Table 2: **Screen vs. surrogate.** Decision utility — *at false-pass tolerance α, the screen avoids X% of human validations* — is the headline scalable-oversight quantity; measuring it on an expanded judge panel is registered as the next analysis and no value is claimed here.

For the frozen experiment we pre-register a smallest effect size of interest (SESOI) of Spearman $\rho_0 = 0.5$ and call an interval containing zero inconclusive rather than negative evidence about the sign. A fitted score mapping $f_{\mathrm{cal}} : \mathbf{P} \mapsto \mathrm{TI}_{m_1}$ is deferred until roughly 40 judge configurations exist; at $n = 5$ primaries the honest object is the pre-specified aggregate with its raw correlation and discrete judge-bootstrap uncertainty.

6 EXPERIMENTS: FROZEN PTI-V0 AND HISTORICAL CONTEXT

6.1 DATA AND HUMAN-ANNOTATION PROTOCOL (AUDITED)

All experiments use VERIBENCH (Miranda et al., 2026), which comprises 857 theorems over five subsets: **HumanEval** (387 theorems from 56 programming puzzles; Chen et al., 2021), **Easy** (278 from 41 introductory tasks), **CS** (68 from 10 classical algorithm problems), **Security** (112 from 28 vulnerability labs), and **RealCode** (12 from 5 Python standard-library programs).

Development panel. Judge validation uses an aligned panel of **75 rated candidate–gold pairs** = 25 problems $\times$ 3 agent generations (generations by Codex-, Claude-, and Llama-family agents); **72 pairs (24 problems) enter all analyses**, because one problem's gold artifact is indexed under a pre-renumbering filename and is skipped by the gold lookup. A repository audit reproduced every published statistic from the raw label files; Appendix A gives the reconciliation, and the supplementary archive contains the tidy panel. **One annotator, two rubric-strictness rounds**: a single expert annotator scored every pair twice on a 0–5 rubric under a *strict* and a *lenient* reading (150 rated labels; 144 analyzed). We refer to these throughout as the strict and lenient views (they were called

`pass1`/`pass2` in earlier versions of this work; those names are inverted relative to chronology, so we avoid temporal language entirely). Inter-round agreement on the 72 analyzed pairs is moderate: Spearman 0.249 (95% confidence interval $[-0.02, 0.49]$), Pearson 0.375, quadratic-weighted $\kappa = 0.372$, MAE 0.771. With one rater, this is *rubric stability*, not inter-rater agreement: the criterion itself moves this much under a strictness change alone, which is why we report strict, lenient, and averaged label views side by side and treat single-view correlations as bounded by this ceiling. **Item selection.** The panel is stratified 5-per-subset across the five VERIBENCH subsets but *hand-picked* within subsets, frozen before any judge experiments, with no selection script or seed and no compile filtering; the estimand is therefore alignment on a fixed curated panel, not on a probability sample of the benchmark (a seeded stratified-selection template exists for a successor study and is the upgrade path if a probability sample is demanded). The same panel is reused across judges, so differences in trend reflect judge behavior rather than row selection.

6.2 PROBE-VALIDITY AUDIT AND EXCLUDED LEGACY DIAGNOSTICS

The development-panel degradation and ALPHA plots used an earlier string-generated probe suite. A judge-free structural re-audit found malformed ALPHA files (including duplicated assignment tokens and inconsistent bound-occurrence renaming) and older degradation ladders that did not satisfy the frozen construction contract. We therefore exclude those plots, their workshop-era 0.399 ALPHA mean, and their associated aggregate comparisons from the evidentiary chain; they are neither reproduced here nor relabeled as $\mathrm{PTI}_{\mathrm{v0}}$. The v5 experiment instead uses repaired, hash-bound manifests whose original and transformed Lean files elaborate under the same toolchain. Invalid constructions are excluded before scoring, while a structurally valid constant-score ladder remains a behavioral failure with factor zero. The validated probe counts, coverage, and property estimates appear in Table 3.

6.3 FROZEN PTI-V0 PROPERTIES AND PRE-SPECIFIED OUTER BRIDGE

Execution and audit. The exact v4 model IDs were unavailable through the approved ChatGPT-backed command-line client, so that panel is `NOT_RUN`; we versioned only the execution/model panel as v5 without changing Eq. 9, the probes, criterion, or analysis policy. The complete run used locally authenticated subscription CLIs with requested/scoring models `gpt-5.6-terra` and `claude-sonnet-5`: 774/774 CLI tasks, 4,728/4,728 unique score cells, and 14,184 bound score records (three per cell), with zero provider-API spend. Claude telemetry also reports a small auxiliary `claude-haiku-4-5` helper on all 258 accepted Claude CLI tasks; we therefore do not claim Sonnet-only execution or prove that the helper could not affect execution. Codex telemetry exposes no served-model field, so the `gpt-5.6-terra` identity rests on the hash-bound explicit model argument and the client's rejection of unsupported identifiers rather than on observed telemetry. Unrelated cases shared a deterministic batch context, but artifact-disjoint packing prevented related levels or alpha partners of the same typed artifact from appearing together. A fresh read-only audit independently reconstructed every score binding, property, criterion statistic, and bootstrap output; all numerical checks passed, with the auxiliary-model provenance recorded as a major limitation.

Coverage and properties. The frozen structural sets contain 86 equality, 49 bug-ladder, 63 missing-specification-ladder, and 67 alpha probes. Every judge has 100% required score coverage and all four properties are `COMPUTED`. Table 3 gives the exact-zero headline point estimates with adjacent uncertainty. Four of five primary configurations have point estimate $\mathrm{PTI}_{\mathrm{v0}} = 0$; independent reconstruction verified at least one directly observed zero artifact factor behind every zero aggregate, rather than missingness or failed parsing. Sonnet–P8 has $\mathrm{PTI}_{\mathrm{v0}} = 0.811$ (95% artifact-cluster bootstrap interval $[0.782, 0.839]$).

Outer bridge and sensitivities. Figure 1 shows the pre-registered primaries-only analysis. Spearman is -0.3536 and Kendall $\tau_b = -0.3162$; the 95% judge-bootstrap interval $[-1.0000, 0.3953]$ contains zero and its upper endpoint lies below the frozen 0.5 SESOI. The explicitly labeled three-property sensitivity $\mathrm{GM}(P_{\mathrm{eq}}, P_{\mathrm{bm}}, P_{\mathrm{ms}})$ has the same rank association because it retains the same four-zero/one-positive tie pattern. Including controls is secondary: Spearman -0.2474, Kendall $\tau_b = -0.2143$, interval $[-0.8165, 0.2568]$, $n = 8$ (6,521/10,000 valid). Leave-one-family-out is not computable: removing Sonnet leaves a constant Terra $\mathrm{PTI}_{\mathrm{v0}}$ vector, while removing Terra leaves only two judges. Exploratory per-property Spearman values are -0.354 (P_{eq}), -0.224 (P_{bm}),

Judge–prompt	Role	P_{eq}	P_{bm}	P_{ms}	$P_{\text{eq}\alpha}$	PTI_{v0}	95% CI
Terra–P8	primary	0.000	0.000	0.000	0.000	0.000	[0, 0]
Terra–P4	primary	0.000	0.000	0.000	0.883	0.000	[0, 0]
Terra–P1	primary	0.000	0.000	0.000	0.939	0.000	[0, 0]
Sonnet–P8	primary	0.919	0.647	0.779	0.937	**0.811**	[.782, .839]
Sonnet–P4	primary	0.000	0.665	0.807	0.904	0.000	[0, .847]
String similarity	control	1.000	0.000	0.846	0.936	0.000	[0, 0]
Edit distance	control	1.000	0.000	0.000	0.999	0.000	[0, 0]
Deterministic random	control	0.424	0.000	0.000	1.000	0.000	[0, 0]

Table 3: **Frozen PTI_{v0} property table.** Values are point estimates; the last column gives artifact-cluster percentile intervals for PTI_{v0}. Exact zeros are observed, not missing-value substitutes; Sonnet–P4's wide interval shows why the four zero point estimates are not equally stable. Per-property intervals appear in Appendix E.

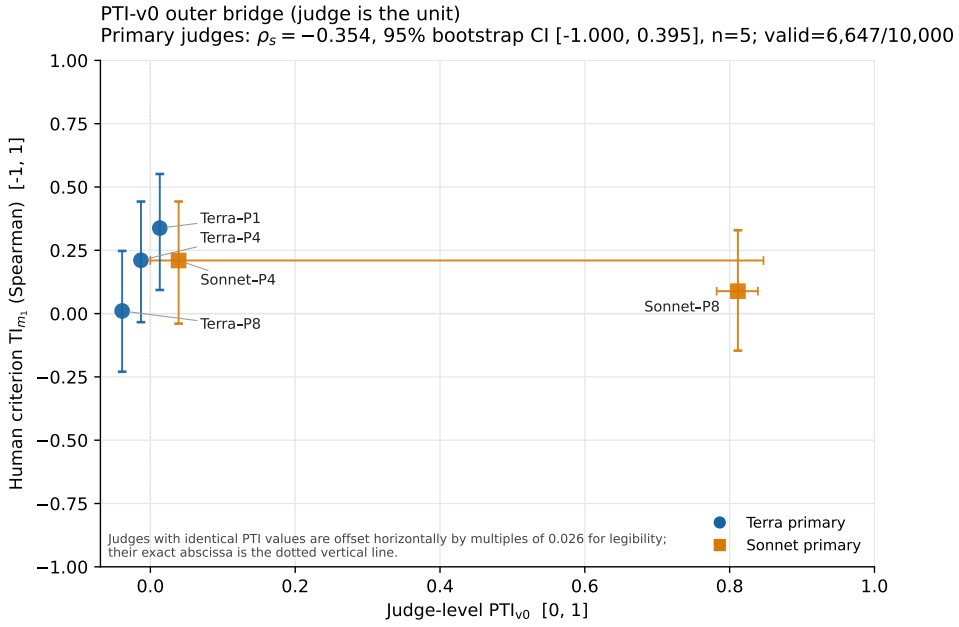


Figure 1: **Outer bridge, one point per primary judge configuration.** Horizontal and vertical bars are 95% artifact-cluster PTI_{v0} and paired-item TI_{m_1} intervals. The four primaries tied at exactly $\text{PTI}_{\text{v0}} = 0$ are offset horizontally by ± 0.013 and ± 0.039 for legibility; their exact abscissa is the dotted vertical line at 0. The bridge statistic resamples judges, not items: Spearman -0.354, 95% bootstrap interval $[-1.000, 0.395]$, $n = 5$ (6,647/10,000 valid replicates). The axes are not a shared measurement scale, so there is no identity line or level-error statistic.

-0.224 (P_{ms}), and 0.600 ($P_{\text{eq}\alpha}$); every interval includes zero, and none is used to redesign the metric post hoc.

SESOI and decision interpretation. The frozen Gaussian-copula simulation fixes the five empirical primary criterion ranks and applies the complete positive rule (point Spearman ≥ 0.5 and lower bootstrap bound > 0). At true Spearman 0.5, only 27.1% of 10,000 simulations pass; the mean realized correlation is 0.475. Thus the panel is low-power, but the observed point and upper interval are also below the SESOI. The correct outcome is *below-SESOI and inconclusive about zero*: no positive bridge, PASS-SCREEN, deployment threshold, causal interpretation, or human-validations-saved claim is supported. The item-level $\text{TI}_{2'}$ diagnostic is statistically distinct from this outer bridge and is shown for a representative judge in Appendix B; its remaining panels ship as supplementary material.

Historical context. Development-panel PTI_{var} associations (including the earlier $0.833/0.905/0.738$ Codex label-view correlations) used a tuned legacy aggregate and are

LEGACY—NOT $\mathrm{PTI}_{\mathrm{v0}}$. They remain useful diagnostics in Appendix F, but they cannot be renamed, pooled with, or used to override this frozen result.

7 CRITERION DRIFT AND THE PRE-REGISTERED HELD-OUT OUTCOME

Separate the frozen result from historical fitting and selection. The $\mathrm{PTI}_{\mathrm{v0}}$ judge panel in Section 6.3 was frozen and its property table committed before the bridge was revealed, so it is the paper's new pre-specified judge-level validation analysis on the 72-row criterion. It is not an out-of-sample or held-out generalization result; by contrast, the development-panel $\mathrm{PTI}_{\mathrm{var}}$ analyses reused human labels while choosing aggregate forms, repeat aggregation, thresholds, and prompt panels, so those are historical validation findings rather than $\mathrm{PTI}_{\mathrm{v0}}$ evidence.

The held-out pool measures a different criterion. Beyond the development panel, an independent pool was collected as the intended held-out test: five raters, 343 scored labels over 78 distinct candidate–gold pairs, many pairs with 4–5 independent ratings. The pool was annotated under a *revised* rubric protocol, and its labels are nearly uncorrelated with the original development labels on shared items (Spearman ≈ 0.13 on 63 shared pairs). A bridge fitted to criterion A and tested on criterion B with $\mathrm{Corr}(A, B) \approx 0.13$ fails at any judge count, so the direct frozen transfer test cannot answer the transfer question, and we withhold its correlations as uninformative rather than negative. *Criterion drift precedes power.* The pool itself is reliable within its own protocol: its estimated pool-mean reliability ceiling is ≈ 0.87, and an annotator re-labeling under the revised protocol matches the pool mean at Spearman 0.90 — label noise cannot be blamed for a null in a protocol-consistent rerun.

Pre-registered screens. Two pre-registered analyses asked whether any label-free property signal predicts judge-level alignment with the revised-protocol criterion. The first (14 configurations) returned its registered "underpowered" outcome. The second — expanded to 32 judge configurations, with candidates, decision thresholds, and analysis seed committed before the new judge scores existed — found no candidate meeting the registered success bar (strongest candidate $|\rho| = 0.39$, Holm-corrected $p = 0.25$; leave-one-out selection transfer $\rho = 0.24$, permutation $p = 0.49$): a registered negative for label-free transfer at this power. The new $\mathrm{PTI}_{\mathrm{v0}}$ experiment asks a narrower question on the original 72-row criterion with a different, frozen five-primary panel; its below-SESOI/inconclusive result is directionally consistent with caution but is not pooled with either revised-protocol screen.

Both sample sizes, both intervals. Power remains binding on top of drift: the frozen outer correlation has five primaries (eight judges including controls), while each inner criterion correlation runs over 72 paired items. We report both uncertainty levels and the outer valid-replicate count; the fix for the outer constraint is more judge configurations (target ≈ 40), not more items.

What this establishes. A protocol requirement rather than a verdict on the method: one labeling protocol end-to-end — fitting and selection against the revised-protocol criterion before any frozen test — with one rubric and one rater pool per claim, and label-lineage verification alongside the behavioral properties. The rubric is part of the criterion's identity; with $|H| = 1$ the panel criterion is one person's judgment under a stated rubric — legitimate as a *scoped* criterion via the $h_j = (\text{person}, \pi)$ decomposition, but it must be said plainly, and we say it here.

8 FAILURE MODES AND LIMITATIONS

ALPHA-renaming. The workshop-era 0.399 mean is not clean evidence of judge failure because its earlier generator produced malformed transforms; Section 6.2 excludes it. On the structurally validated frozen probes, $P_{\mathrm{eq}\alpha}$ is 0.000, 0.883, and 0.939 for the three Terra prompts and 0.937 and 0.904 for the two Sonnet prompts. Thus the fresh result is configuration-dependent rather than a family-wide failure, although Terra–P8 contains two directly observed zero factors. Whether ALPHA-renaming is human-invariant on these artifacts is still the margin measurement Route B needs (Appendix C); without that measurement, even the valid probe is a behavioral diagnostic rather than proof of semantic misalignment.

Limitations. The judge is a proxy: scores are prompt- and rubric-sensitive and do not constitute kernel-level checking; evaluations are specification-relative (Appendix I). No finite probe suite cer-

tifies all failure modes; untested modes may persist. The development panel is small, single-rater, and hand-curated, so the criterion is scoped to that panel and rubric. The frozen bridge has only five primaries, four point estimates tied at zero, a wide discrete bootstrap interval, and only 27.1% simulated positive-rule power at the SESOI. It substitutes accessible v5 models for an exact v4 panel that was not run; unrelated cases share batch context; and every accepted Claude CLI task records an auxiliary Haiku helper alongside Sonnet. The separate revised-protocol screens returned underpowered and negative outcomes on a drifted criterion — cross-protocol transfer is bounded, not settled. Item-level extrapolation ($b' \to b$: 72 validated items vs. 470 observed typed candidates over 157 items; nominally 471 with one missing candidate) is a stated assumption, not a measured one.

9 Conclusion

We reframed judge validation as measurement: a construct (TW_m), a criterion (TI_m) on a specified panel, and a label-free instrument ($\mathbf{P}$, PTI_p) whose acceptance semantics are a screen backed by a bridge. The exact frozen $\mathrm{PTI}_{\mathrm{v0}}$ experiment does not support its positive bridge: four of five primary point estimates are zero and the judge-level association is below-SESOI/inconclusive. Historical development-panel associations are not $\mathrm{PTI}_{\mathrm{v0}}$ evidence; on a drifted criterion a separate pre-registered screen likewise found no qualifying signal, and auditing the labels showed how much of "ground truth" is protocol. The next steps are mechanical: one rubric end-to-end, $\approx$40 judge configurations for a powered bridge, CLI/model provenance that excludes auxiliary execution, per-property screening bounds frozen before data, Route B margin measurement, and decision utility as the headline oversight quantity.

Ethics Statement

This work reuses research-team annotations of machine-generated Lean artifacts and contains no personal or sensitive attributes. We disclose the single-rater scope and rubric instability rather than treating the labels as universal ground truth. Over-reliance on an imperfect proxy judge could misrank systems or obscure specification errors; accordingly, we frame PTI as a diagnostic screen, not a replacement for human review or kernel-level validation.

Reproducibility Statement

The supplementary material is assembled from an explicit anonymized allowlist and contains the tidy development panel (75 rated / 72 analyzed pairs with strict, lenient, and averaged views), structural manifests for 470 observed typed candidates, the frozen $\mathrm{PTI}_{\mathrm{v0}}$ config and score-request bindings, all 14,184 normalized score records and identity-scanned CLI transcripts, per-ladder diagnostics, property/criterion/bridge/SESOI JSON and CSV authorities, figure source data, and the pre-registration documents for the separate held-out screens. The canonical commands use metric version `PTI_v0`, 10,000 replicates, seed 0, and a content-bound pre-bridge property authority identified inside the anonymized bundle. Appendix A reconciles the human counts; every new manuscript number is mapped to a machine-readable artifact in the supplementary result-status table.

AI Use Statement

Large language models were used in two capacities. (i) As research subjects: all judges under study are LLMs, as described throughout. (ii) As assistants: LLM-based coding tools helped implement and test experiment and analysis code, execute the prescribed workflow, audit the repository, cross-check quantitative outputs, summarize and interpret results, and draft and edit manuscript text and LaTeX. No LLM output was accepted as a substitute for the frozen statistical artifacts or deterministic checks; AI-assisted code and quantitative claims were tested or independently checked against committed artifacts. The authors take full responsibility for the final content.

REFERENCES

Riyaz Ahuja, Jeremy Avigad, Prasad Tetali, and Sean Welleck. Improver: Lean-aware chain-of-states prompting for proof optimization, 2024. URL https://arxiv.org/abs/2410.04753.

Leni Aniva, Chuyue Sun, Brando Miranda, Clark Barrett, and Sanmi Koyejo. Pantograph: A machine-to-machine interaction interface for advanced theorem proving, high level reasoning, and data extraction in lean 4, 2024. URL https://arxiv.org/abs/2410.16429.

Anonymous. Trusting the formal judge: Certifying LLM theorem scorers for code verification in lean 4. *NeurIPS (to appear)*, 2025.

Randy Bach et al. A systematic literature review of user trust in AI-enabled systems: An HCI perspective. *International Journal of Human–Computer Interaction*, 2022.

Anna Bavaresco, Raffaella Bernardi, Leonardo Bertolazzi, Desmond Elliott, Raquel Fernández, Albert Gatt, Esam Ghaleb, Mario Giulianelli, Michael Hanna, Alexander Koller, André F. T. Martins, Philipp Mondorf, Vera Neplenbroek, Sandro Pezzelle, Barbara Plank, David Schlangen, Alessandro Suglia, Aditya K. Surikuchi, Ece Takmaz, and Alberto Testoni. Llms instead of human judges? a large scale empirical study across 20 nlp evaluation tasks, 2024. URL https://arxiv.org/abs/2406.18403.

Samuel R Bowman et al. Measuring progress on scalable oversight for large language models. In *arXiv preprint arXiv:2211.03540*, 2022.

Sergiu Bursuc, Theodore Ehrenborg, Shaowei Lin, Lacramioara Astefanoaei, Ionel Emilian Chiosa, Jure Kukovec, Alok Singh, Oliver Butterley, Adem Bizid, Quinn Dougherty, Miranda Zhao, Max Tan, and Max Tegmark. A benchmark for vericoding: formally verified program synthesis, 2025. URL https://arxiv.org/abs/2509.22908.

Zheng Cheng Chan et al. Chateval: Towards better llm-based evaluators through multi-agent debate, 2024. URL https://arxiv.org/abs/2308.07201.

Eric Chen, Aryan Gulati, Brando Miranda, Zeyu Tang, and Sanmi Koyejo. Rethinking LLM judges: Chain-of-thought and multi-step pipelines for math grading. In *ICLR 2026 Workshop on Logical Reasoning of Large Language Models*, 2026. URL https://openreview.net/forum?id=vdXPorr099.

Mark Chen et al. Evaluating large language models trained on code, 2021. URL https://arxiv.org/abs/2107.03374.

Ching-An Cheng, Allen Nie, and Adith Swaminathan. Trace is the next autodiff: Generative optimization with rich feedback, execution traces, and llms, 2024. URL https://arxiv.org/abs/2406.16218.

Yegor Denisov-Blanch, Joshua Kazdan, Jessica Chudnovsky, Rylan Schaeffer, Sheng Guan, Soji Adeshina, and Sanmi Koyejo. Consensus is not verification: Why crowd wisdom strategies fail for LLM truthfulness, 2026. URL https://arxiv.org/abs/2603.06612.

Z. Ding et al. Citations and trust in LLM-generated responses. In *AAAI*, 2025.

Mara Finkelstein, Daniel Deutsch, Parker Riley, Juraj Juraska, Geza Kovacs, and Markus Freitag. From jack of all trades to master of one: Specializing LLM-based autoraters to a test set. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pp. 17221–17238. PMLR, 2025. URL https://proceedings.mlr.press/v267/finkelstein25a.html.

Lucas Torroba Hennigen, Zejiang Shen, Aniruddha Nrusimha, Bernhard Gapp, David Sontag, and Yoon Kim. Towards verifiable text generation with symbolic references. In *First Conference on Language Modeling*, 2024. URL https://openreview.net/forum?id=fib9qidCpY.

Zachary Kenton, Noah Y Siegel, János Kramár, Jonah Brown-Cohen, Samuel Albanie, Jannis Bulian, Rishabh Agarwal, David Lindner, Yunhao Tang, Noah D Goodman, and Rohin Shah. Scalable oversight with weak LLMs judging strong LLMs. In *Proceedings of the 38th International Conference on Neural Information Processing Systems (NeurIPS)*, 2024. URL https://arxiv.org/abs/2407.04622.

Omar Khattab, Arnav Singhvi, Paridhi Maheshwari, Zhiyuan Zhang, Keshav Santhanam, Sri Vardhamanan, Saiful Haq, Ashutosh Sharma, Thomas T. Joshi, Hanna Moazam, Heather Miller, Matei Zaharia, and Christopher Potts. Dspy: Compiling declarative language model calls into self-improving pipelines, 2023. URL https://arxiv.org/abs/2310.03714.

Shulei Li et al. Can you trust llm judgments? reliability of llm-as-a-judge, 2024. URL https://arxiv.org/abs/2412.12509.

Chengwu Liu, Ye Yuan, Yichun Yin, Yan Xu, Xin Xu, Zaoyu Chen, Yasheng Wang, Lifeng Shang, Qun Liu, and Ming Zhang. Safe: Enhancing mathematical reasoning in large language models via retrospective step-aware formal verification, 2025. URL https://arxiv.org/abs/2506.04592. Introduces the FormalStep benchmark for step-level verification in Lean 4.

Yang Liu et al. G-eval: Nlg evaluation using gpt-4 with better human alignment, 2023. URL https://arxiv.org/abs/2303.16634.

Qianou Ma, Hua Shen, Kenneth Koedinger, and Tongshuang Wu. How to teach programming in the ai era? using llms as a teachable agent for debugging, 2024. URL https://arxiv.org/abs/2310.05292.

Brando Miranda, Srivatsava Daruru, Ethan S Hersch, Zhanke Zhou, Allen Nie, Daneshvar Amrollahi, Leni Aniva, Iddah Mlauzi, Kirill Acharya, Elyas Obbad, Dilara Soylu, Weston Kirk, Zixiao Jolene Wang, Kai Fronsdal, Ying Li, Donald Poindexter Jr, Rakshit Kaushik, Shurui Liu, Yegor Denisov-Blanch, Steven Dillmann, Simon Obstbaum, Santiago Cuellar, John Sarracino, Rylan Schaeffer, Mo Tiwari, Donghyun Lee, Bo Han, and Sanmi Koyejo. Veribench: An end-to-end formal verification benchmark for AI coding agents in lean 4. In *3rd AI for Math Workshop: Toward Self-Evolving Scientific Agents*, 2026. URL https://openreview.net/forum?id=lkL0qnUv3p.

A. Naik et al. TiCoder: LLM-based test-driven interactive code generation. *IEEE Transactions on Software Engineering*, 2024.

Fan Nie, Ken Ziyu Liu, Zihao Wang, Rui Sun, Wei Liu, Weijia Shi, Huaxiu Yao, Linjun Zhang, Andrew Y. Ng, James Zou, Sanmi Koyejo, Yejin Choi, Percy Liang, and Niklas Muennighoff. Uq: Assessing language models on unsolved questions, 2025. URL https://arxiv.org/abs/2508.17580.

Ospanov et al. Apollo: Recursive repair with lean feedback boosts minif2f proof success. arXiv preprint, 2023.

H. G. Rice. Classes of recursively enumerable sets and their decision problems. *Transactions of the American Mathematical Society*, 74(2):358–366, 1953. doi: 10.2307/1990888. URL https://doi.org/10.2307/1990888.

Zachary Robertson and Sanmi Koyejo. Let's measure information step-by-step: AI-based evaluation beyond vibes, 2025. URL https://arxiv.org/abs/2508.05469.

Olawale Salaudeen and Moritz Hardt. Imagenot: A contrast with imagenet preserves model rankings, 2024. URL https://arxiv.org/abs/2404.02112.

Olawale Salaudeen, Anka Reuel, Ahmed Ahmed, Suhana Bedi, Zachary Robertson, Sudharsan Sundar, Ben Domingue, Angelina Wang, and Sanmi Koyejo. Measurement to meaning: A validity-centered framework for ai evaluation, 2025. URL https://arxiv.org/abs/2505.10573.

Chuyue Sun, Ying Sheng, Oded Padon, and Clark Barrett. Clover: Closed-loop verifiable code generation, 2024. URL https://arxiv.org/abs/2310.17807.

Zhengyang Tang, Ziniu Li, Zhenyang Xiao, Tian Ding, Ruoyu Sun, Benyou Wang, Dayiheng Liu, Fei Huang, Tianyu Liu, Bowen Yu, and Junyang Lin. Self-evolving critique abilities in large language models, 2025. URL https://arxiv.org/abs/2501.05727.

Yanfei Tong and Chiyuan Zhang. Codejudge: Evaluating code generation with large language models, 2024. URL https://arxiv.org/abs/2410.02184.

Ruida Wang, Jipeng Zhang, Yizhen Jia, Rui Pan, Shizhe Diao, Renjie Pi, and Tong Zhang. Theoremllama: Transforming general-purpose llms into lean4 experts, 2024a. URL https://arxiv.org/abs/2407.03203.

Ruotong Wang, Ruijia Cheng, Denae Ford, and Thomas Zimmermann. Investigating and designing for trust in ai-powered code generation tools. In *The 2024 ACM Conference on Fairness, Accountability, and Transparency*, FAccT '24, pp. 1475–1493. ACM, June 2024b. doi: 10.1145/3630106.3658984. URL http://dx.doi.org/10.1145/3630106.3658984.

Tongshuang Wu, Ellen Jiang, Aaron Donsbach, Jeff Gray, Alejandra Molina, Michael Terry, and Carrie J Cai. Promptchainer: Chaining large language model prompts through visual programming, 2022. URL https://arxiv.org/abs/2203.06566.

Hao Yan, Thomas D. Latoza, and Ziyu Yao. Intelliexplain: Enhancing conversational code generation for non-professional programmers, 2024. URL https://arxiv.org/abs/2405.10250.

Lan Zhang, Marco Valentino, and Andre Freitas. Beyond gold standards: Epistemic ensemble of llm judges for formal mathematical reasoning, 2025. URL https://arxiv.org/abs/2506.10903.

Lianmin Zheng et al. Judging llm-as-a-judge with mt-bench and chatbot arena, 2023. URL https://arxiv.org/abs/2306.05685.

Jin Peng Zhou, Charles Staats, Wenda Li, Christian Szegedy, Kilian Q. Weinberger, and Yuhuai Wu. Don't trust: Verify – grounding llm quantitative reasoning with autoformalization, 2024. URL https://arxiv.org/abs/2403.18120.

Jin Peng Zhou, Sébastien M. R. Arnold, Nan Ding, Kilian Q. Weinberger, Nan Hua, and Fei Sha. Graders should cheat: privileged information enables expert-level automated evaluations, 2025. URL https://arxiv.org/abs/2502.10961.

Terry Yue Zhuo. ICE-score: Instructing large language models to evaluate code. In Yvette Graham and Matthew Purver (eds.), *Findings of the Association for Computational Linguistics: EACL 2024*, pp. 2232–2242, St. Julian's, Malta, March 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.findings-eacl.148. URL https://aclanthology.org/2024.findings-eacl.148/.

A AUDITED DATA FACTS AND COUNTS RECONCILIATION

This appendix states the audited facts about the human-annotation data that the main text relies on; the audit scripts and tidy panel ship in the supplementary archive.

Panel composition. The label files hold **75 rated candidate–gold pairs over 25 problems** (25 problems $\times$ 3 agent generations; 5 problems per VERIBENCH subset $\times$ 5 subsets $\times$ 3 generations $= 75$). Every published judge-alignment and inter-round number is computed on **72 pairs over 24 problems**: one problem's rows carry a pre-renumbering filename, the gold lookup misses, and its 3 pairs are skipped as `missing_gold`. The dropped rows are a stale *key*, not missing data — the candidate artifacts exist on disk under the renumbered name — so the two repair options are (a) this paper's wording (75 rated / 72 analyzed) or (b) renaming the 3 label rows and regenerating all dependent statistics; we adopt (a) and register (b) as a data-repair task.

The two unrelated "3"s. The 75-vs-72 gap (3 dropped pairs) is distinct from a second legacy artifact: in the *lenient* label file one problem appears twice (all 3 of its candidates, scored 1 and 2). Those duplicate rows collapse to the same 75 pairs and never inflated the panel; averaging them produces the three 1.5 scores visible in Figure 7.

Label totals and views. One expert annotator, two rubric-strictness rounds: 150 rated labels (75×2), 144 analyzed (72×2), on a 0–5 rubric, plus a derived averaged view. The strict and lenient names replace the earlier `pass1`/`pass2`, whose numbering is inverted relative to chronology (verified by content hash against the raw files); no per-row rater field exists anywhere in the label data, so rater identity is carried by file provenance only.

Inter-round agreement, one convention. On the 72 analyzed pairs, with duplicate rows averaged before comparison (mean-of-duplicates, the same convention as the averaged view): Spearman 0.249 (95% CI $[-0.015, 0.489]$), Pearson 0.375 ($[0.076, 0.604]$), Kendall τ 0.221, quadratic-weighted κ 0.372 ($[0.073, 0.581]$), MAE 0.771, MSE 1.122. An earlier version reported MAE 0.778 and MSE 1.139, which mix a first-of-duplicates convention into the same table; the values above are self-consistent under one convention. With one rater these statistics are *rubric stability*, not inter-rater agreement.

Item selection provenance. Stratified 5-per-subset, hand-picked within subset, frozen before any judge experiments; no selection script, no seed, no compile filtering. The estimand is alignment on a fixed curated panel. A seeded, audited stratified-selection template (30 problems, fixed seed, per-rater sheets) exists for a successor study and is the upgrade path to a probability sample.

The independent five-rater pool. 343 labels over 78 distinct pairs from 5 raters under a revised rubric; overlap with the development panel is 63 pairs; Spearman between pool mean and original development labels on shared items is ≈ 0.13. The two label populations are not interchangeable, and any experiment that fits or selects on one and tests on the other is confounded by construct drift (Section 7).

Agents on the panel. The three generations per problem come from Codex-, Claude-, and Llama-family agents. The typed pool contains 470 observed candidates over 157 problems; the nominal $157 \times 3 = 471$ grid has one documented missing candidate (problem granularity; the benchmark's 857 counts theorems).

B ITEM-LEVEL CRITERION DIAGNOSTIC

The outer bridge in Figure 1 has one point per judge; the inner diagnostic below has one point per paired item for a single historical judge and explains what TI_{m_1} measures. Marker area counts tied item pairs, vertical bars show within-item judge repeat standard deviation, and the top/right marginals expose range restriction without asserting a paired dependence pattern. There is no identity line because human and judge scores are not mapped to a common measurement scale. The displayed GPT-5.3-Codex configuration is illustrative and predates the v5 $\mathrm{PTI}_{\mathrm{v0}}$ panel; its $\mathrm{TI}_{2'}$ value is not an outer-bridge point in Experiment 29.

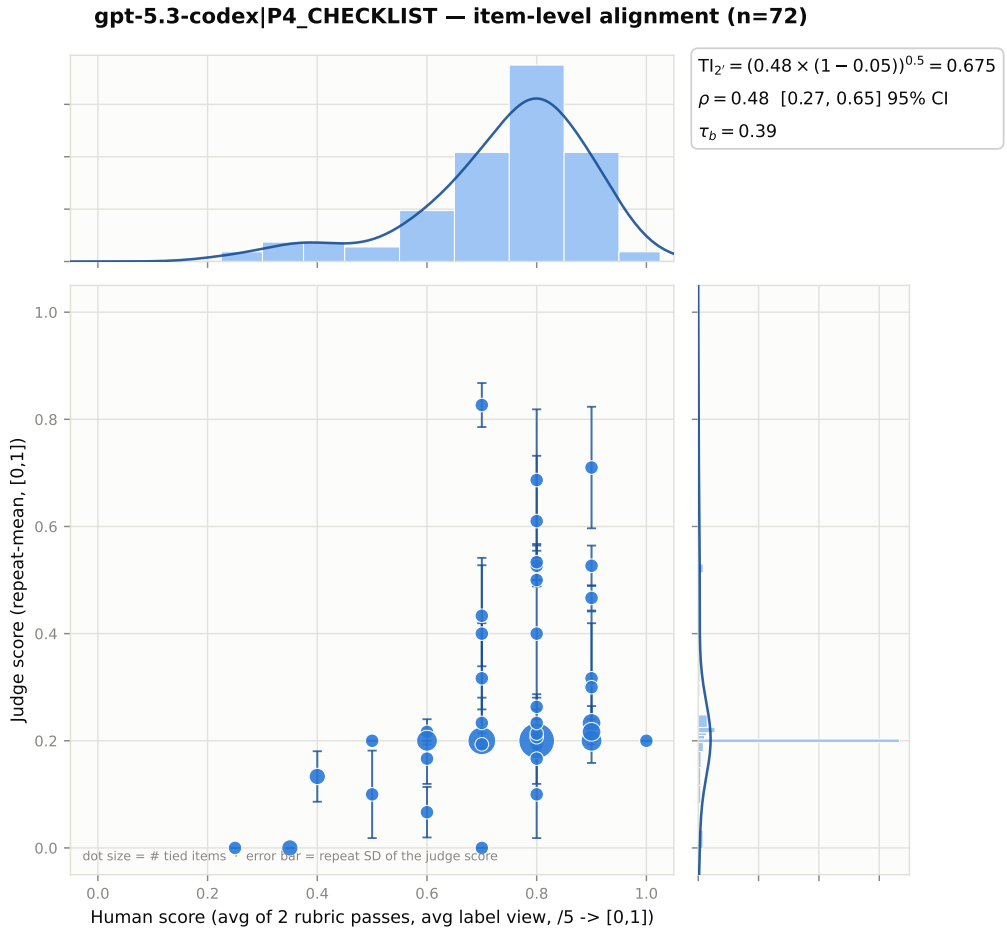


Figure 2: **Inner item-level criterion diagnostic (historical illustrative judge).** Spearman 0.48 (95% paired-item bootstrap interval $[0.27, 0.65]$), Kendall $\tau_b = 0.39$, and repeat-instability 0.05 yield $\mathrm{TI}_{2'} = 0.675$ over 72 paired items. This diagnostic decomposes criterion validity and within-item repeat stability; it is statistically distinct from the judge-level $\mathrm{PTI}_{\mathrm{v0}}$ bridge.

C Route B: Conditional Falsification Lemma

Lemma 1 (Conditional falsification)**.** *Suppose the judge is pointwise ε-aligned on a class Z: $|J(z) - H(z)| \leq \varepsilon$ for all $z \in Z$. Let T be a transformation with human margin $H(x) - H(Tx) \geq \delta$ for $x, Tx \in Z$. Then $J(x) - J(Tx) \geq \delta - 2\varepsilon$. Consequently, an observed reversal $J(Tx) - J(x) > 2\varepsilon - \delta$ refutes pointwise ε-alignment on Z; and if T is human-invariant ($\delta = 0$ with equality), then $|J(x) - J(Tx)| \leq 2\varepsilon$.*

Proof. $J(x) \geq H(x) - \varepsilon \geq H(Tx) + \delta - \varepsilon \geq J(Tx) + \delta - 2\varepsilon$. □

The paper-level sentence licensed by this lemma: *under stated metamorphic-validity and margin assumptions, sufficiently large probe violations falsify pointwise ε-alignment.* The margins δ are currently assumed, not measured. A small human collection on perturbed ladder items (a few dozen ratings on bug-ladder rungs and ALPHA pairs) would (i) ground the lemma's premises empirically and (ii) adjudicate whether ALPHA-renaming is human-invariant on these artifacts, which is the standing P_4 failure mode. This is registered as the δ-measurement collection.

D HISTORICAL AGGREGATE DEFINITIONS (LEGACY—NOT PTI-V0)

Every aggregate and association in this section predates the frozen $\mathrm{PTI}_{\mathrm{v0}}$ experiment. They are retained for transparency about earlier fitting and selection and must not be renamed, pooled with, or interpreted as $\mathrm{PTI}_{\mathrm{v0}}$.

The arithmetic baselines are

$$v1 = 0.30\,\text{ident} + 0.30\,\text{mono_rank} + 0.25\,\text{weak} + 0.15\,\text{stab},$$

$$v2 = 0.20\,\text{ident} + 0.20\,\text{mono_rank} + 0.15\,\text{weak} + 0.10\,\text{stab} + 0.20\,\text{anti} + 0.15\,\text{subtle},$$

where mono_rank $= (1 - \rho_{\text{bug}})/2$ rescales bug-ladder Spearman so larger is better, and stab $=$ $1 -$ flip_rate. For the legacy variance-aware aggregate, the quality term is

$$C_f = (\text{ident} \cdot \text{mono_rank} \cdot \text{weak} \cdot \text{stab})^{1/4},$$

and $\widetilde{\mathrm{VarCorr}}, \widetilde{\mathrm{VarScore}}$ are normalized variance penalties (lower is better before $1 - \cdot$). The plain geometric comparator used in the historical analyses is $\mathrm{GM} = \mathrm{mono}^{0.4} \cdot \mathrm{stab}^{0.6}$.

Leanstral negative control under re-aggregation. To stress-test whether aggregate choice alone can rescue a weak judge family, we re-aggregate the same Leanstral judge outputs under AM ($v2$), GM, and $\mathrm{PTI}_{\mathrm{var}}$:

Label view	AM (v2)	GM	$\mathrm{PTI}_{\mathrm{var}}$
strict	-0.5952	-0.1429	-0.1190
lenient	-0.1905	-0.2619	-0.0476
avg	-0.4286	-0.2143	-0.0952

Table 4: **Leanstral remains low-alignment under multiple aggregate constructions.** Re-aggregation improves over the arithmetic baseline in some views, but all settings remain near-zero or negative, suggesting the bottleneck is the underlying judge behavior for this pairwise-rubric task rather than aggregate form. One testable conjecture: Leanstral is stronger at Lean proof editing/generation than at pairwise judging assessed against human ratings under this rubric.

E PER-PROPERTY DIAGNOSTICS AND FAILURE MODES

Excluded legacy suite. The earlier low-variance replot package is intentionally absent: its source variants predate the judge-free structural gate, and the ALPHA generator used for the workshop-era result was malformed. Those plots remain repository history only and cannot establish monotonicity or invariance.

Fresh exact-zero diagnoses. Experiment 29 retains every structurally valid constant-score ladder as factor zero. Each of the four zero-valued primary $\mathrm{PTI}_{\mathrm{v0}}$ point estimates has at least one directly observed zero artifact factor in the committed property authority; no missing or unparsable score is converted to zero. Terra–P8 has zeros in all four properties; Terra–P4 and Terra–P1 have nonzero $P_{\mathrm{eq}\alpha}$ but zeros in the other properties; Sonnet–P4 has nonzero degradation and ALPHA factors but a zero equality factor.

Judge–prompt	P_{eq}	P_{bm}	P_{ms}	$P_{\mathrm{eq}\alpha}$	$\mathrm{PTI}_{\mathrm{v0}}$
Terra–P8	[0, 1]	[0, 0]	[0, 0]	[0, .926]	[0, 0]
Terra–P4	[0, 0]	[0, 0]	[0, 0]	[.788, .963]	[0, 0]
Terra–P1	[0, 0]	[0, 0]	[0, 0]	[.867, .983]	[0, 0]
Sonnet–P8	[.842, .982]	[.574, .714]	[.748, .807]	[.918, .953]	[.782, .839]
Sonnet–P4	[0, .999]	[.612, .717]	[.781, .832]	[.864, .933]	[0, .847]
String control	[1, 1]	[0, 0]	[.804, .880]	[.883, .975]	[0, 0]
Edit distance	[1, 1]	[0, 0]	[0, 0]	[.999, 1]	[0, 0]
Random	[.359, .495]	[0, 0]	[0, 0]	[1, 1]	[0, 0]

Table 5: **Artifact-cluster 95% percentile intervals for every frozen property and $\mathrm{PTI}_{\mathrm{v0}}$.** A zero point estimate can have a nonzero upper endpoint when a bootstrap resample omits the artifact carrying the exact-zero factor; this is most consequential for Sonnet–P4.

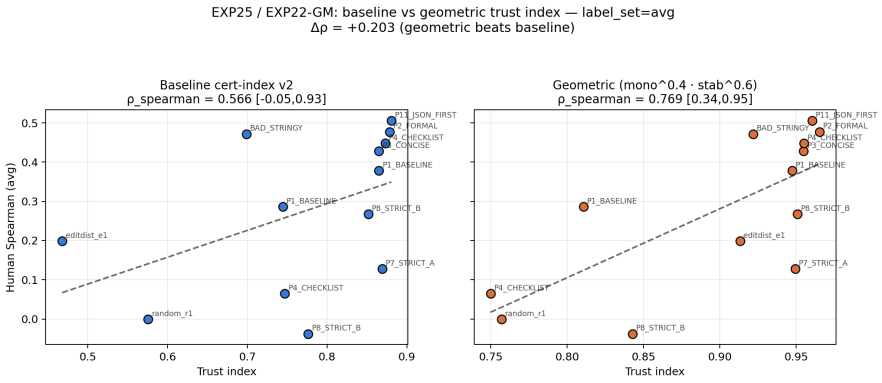
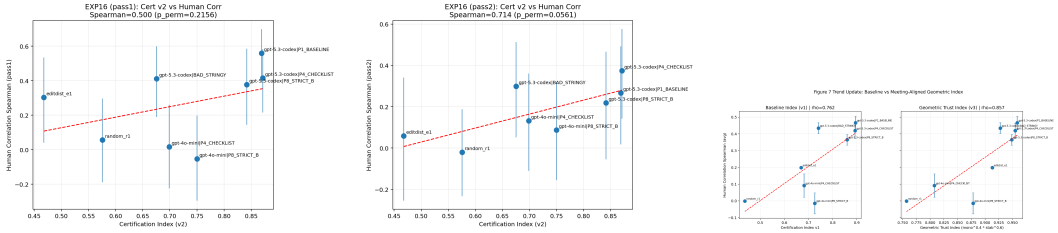


Figure 3: **Expanded 13-judge panel trend (avg validation labels).** The geometric aggregate retains strong alignment with TI_{m_1} at larger judge-panel size ($\rho = 0.769$); view-specific values are strict $\rho = 0.764$, lenient $\rho = 0.753$.



(a) Judge-level trend (strict) (b) Judge-level trend (lenient) (c) Judge-level trend (avg)

Figure 4: **Judge-level aggregate-to-criterion trend across label views.** Higher aggregate property satisfaction is associated with stronger human agreement in every view.

Fresh ALPHA diagnostic. Across the five primaries, validated $P_{\mathrm{eq}\alpha}$ ranges from 0 to 0.939. The broad spread motivates a future artifact-level analysis, but it is not selected post hoc as an alternative headline metric; all per-property bridge intervals include zero.

F HISTORICAL VALIDATION-SET BRIDGE (LEGACY—NOT PTI-V0)

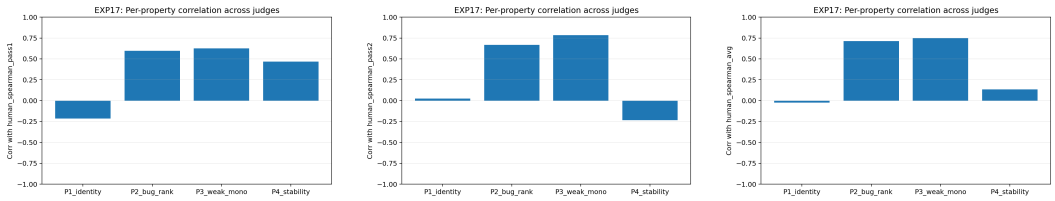
LEGACY—NOT $\mathrm{PTI}_{\mathrm{v0}}$. This appendix reports full visual diagnostics for the development-panel bridge analyses under the three label views (strict, lenient, avg); several embedded figure legends predate the current notation and show `pass1`/`pass2` (strict = `pass1`, lenient = `pass2`) or the legacy aggregate names "cert index" / "trust index (geometric)" for the quantities now written $v1$/GM/$\mathrm{PTI}_{\mathrm{var}}$. We evaluate eight judge configurations spanning frontier-model prompts, weaker LLM baselines, edit-distance heuristics, and random controls. 10,000-resample bootstrap 95% confidence intervals for the headline bridge coefficients are $[0.24, 1.00]$ (avg), $[0.41, 1.00]$ (strict), and $[0.11, 1.00]$ (lenient), with two-sided permutation p-values 0.017, 0.005, and 0.045 (fixed seed 42); the intervals are wide, as expected at this panel size, but all exclude zero.

F.1 LATEST RERUN (HIGHER REPEATS, ROBUST AGGREGATION)

To reduce judge-call noise, the multi-judge validation trend was rerun with `JUDGE_REPEATS=9`, median repeat aggregation, 10k bootstrap and 20k permutation resamples:

F.2 NAIVE BLIND BASELINE

A single-pass, surface-form-oriented blind rubric without property-based tuning tracks lenient labels somewhat but remains substantially weaker and less consistent than the top-ranked judges:



(a) Per-property trend (strict) (b) Per-property trend (lenient) (c) Per-property trend (avg)

Figure 5: **Property-wise diagnostics.** P3 monotonicity is consistently the strongest predictor across label views; P2 second; P1/P4 weaker standalone at this scale.

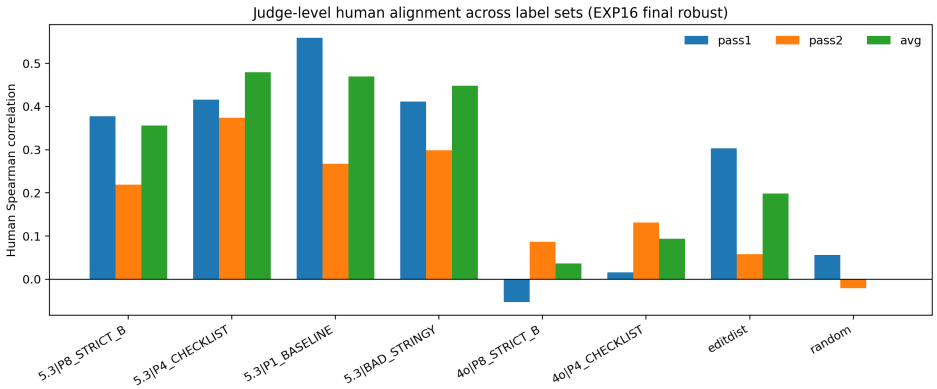


Figure 6: **Judge-wise human agreement across label views.** GPT-5.3-Codex variants remain above weaker baselines on most views; view-to-view variation motivates reporting all three.

F.3 Human-agreement ceiling context

Judge–human agreement should be read against the reliability of the labels themselves. On the 72 analyzed pairs, inter-round agreement is moderate (Table 8), so validation-set judge–human Spearman values in the 0.48–0.56 range approach the available annotation ceiling on this panel rather than being "low absolute" values.

F.4 Cross-model validation status

The Claude Opus family run is a full-panel run of the same form as Codex and Qwen3-Coder: eight configurations (four Opus $\times$ prompts, cross-family `gpt-4o-mini` $\times$ two prompts, edit/random controls), repeated-judge aggregation, identical label views. Early Claude 3.x Sonnet scatterplots were small-file protocol iterations used to stabilize prompts and ladders; they are omitted because they predate the frozen structural gate and are not behind any headline statistic.

G Threshold Selection (Legacy—Not PTI-v0)

A threshold t converts a continuous judge score into a yes/no decision; flip rate measures how often repeated judge calls on the same item disagree after thresholding. Across $t \in \{0.60, 0.65, 0.70, 0.75, 0.80, 0.85\}$, ranking behavior is unchanged ($\rho_{\text{bug}} = -0.790$ throughout), while stricter thresholds modestly reduce flip rate at some coverage cost; $t = 0.85$ gives the lowest observed flip rate with pass rate near 0.5.

Label view	v1 Spearman	v2 Spearman
strict	0.762 ($p = 0.0372$, CI $[-0.139, 1.000]$)	0.524 ($p = 0.1959$, CI $[-0.400, 0.975]$)
lenient	0.810 ($p = 0.0222$, CI $[0.291, 1.000]$)	0.571 ($p = 0.1477$, CI $[-0.231, 0.975]$)
avg	0.762 ($p = 0.0402$, CI $[0.103, 1.000]$)	0.476 ($p = 0.2426$, CI $[-0.425, 1.000]$)

Table 6: **Higher-repeat rerun (9 repeats, median aggregation).** The $v1$ trend is preserved; $v2$ remains more sensitive at $n_{\text{judges}} = 8$.

Label view	ρ_s	r	τ	MAE
strict	0.171 ($p = 0.151$)	0.218 ($p = 0.0656$)	0.155 ($p = 0.144$)	1.049
lenient	0.388 ($p = 7.58 \times 10^{-4}$)	0.427 ($p = 1.87 \times 10^{-4}$)	0.343 ($p = 9.40 \times 10^{-4}$)	1.333

Table 7: **Naive blind baseline vs. validation labels (0–5 scale).** Included as a negative control; not used for ranking claims.

H JUDGE PROMPT TEMPLATES

The primary Codex / Opus / Qwen comparisons use the prompt variants `P1_BASELINE`, `P4_CHECKLIST`, and `P8_STRICT_B`; the `BAD_STRINGY` lexical baseline is a negative control.

P1_BASELINE

```
You are an expert in Lean 4 formal verification.
Score how well the GENERATED THEOREM captures the correctness properties
of the REFERENCE IMPLEMENTATION. Score from 0.0 to 1.0.

- 1.0: Semantically equivalent to what the reference guarantees
- 0.7-0.9: Captures most properties, misses minor ones
- 0.4-0.6: Captures some properties, has meaningful gaps
- 0.1-0.3: Technically valid Lean but misses core properties
- 0.0: Trivially true (e.g., `True`) or incorrect

REFERENCE IMPLEMENTATION:
{reference_impl}

GENERATED THEOREM:
{generated_theorem}

Respond ONLY with valid JSON, no markdown:
{"score": <float 0.0-1.0>,
 "reasoning": "<one sentence max>"}
```

P4_CHECKLIST

```
Evaluate this in order:
1) What properties does REFERENCE IMPLEMENTATION guarantee?
2) Which are preserved by GENERATED THEOREM?
3) Are there missing core guarantees or trivialization?
4) Map to score in [0.0, 1.0]:
   1.0 equivalent; 0.7-0.9 minor gaps;
   0.4-0.6 meaningful gaps; 0.1-0.3 weak;
   0.0 trivial/incorrect.

REFERENCE IMPLEMENTATION:
{reference_impl}

GENERATED THEOREM:
{generated_theorem}

Return JSON only:
{"score": <float 0.0-1.0>,
 "reasoning": "<one sentence>"}
```

P8_STRICT_B

```
Evaluate conservatively. If uncertain, choose the lower score.
Use [0.0,1.0]:
1.0 equivalent semantics
0.7-0.9 mostly right, minor misses
0.4-0.6 substantial omissions
```

Quantity	Value	Notes
Inter-round Spearman (strict vs lenient)	0.249	95% CI = $[-0.015, 0.489]$
Inter-round Pearson (strict vs lenient)	0.375	95% CI = $[0.076, 0.604]$
Inter-round Kendall τ	0.221	analyzed $n = 72$
Inter-round QWK	0.372	95% CI = $[0.073, 0.581]$
Inter-round MAE	0.771	analyzed $n = 72$, 0–5 rubric, mean-of-duplicates
Inter-round MSE	1.122	analyzed $n = 72$, 0–5 rubric, mean-of-duplicates
Best validation-set Spearman (avg labels)	0.480	best single judge, avg view
Best validation-set Spearman (strict labels)	0.559	best single judge, strict view

Table 8: **Single-rater rubric-stability context (analyzed $n = 72$).** All inter-round statistics use one duplicate-handling convention (mean-of-duplicates); an earlier version mixed conventions within this table (MAE 0.778, MSE 1.139), corrected here. With one annotator, these are rubric-stability figures, not inter-rater agreement.

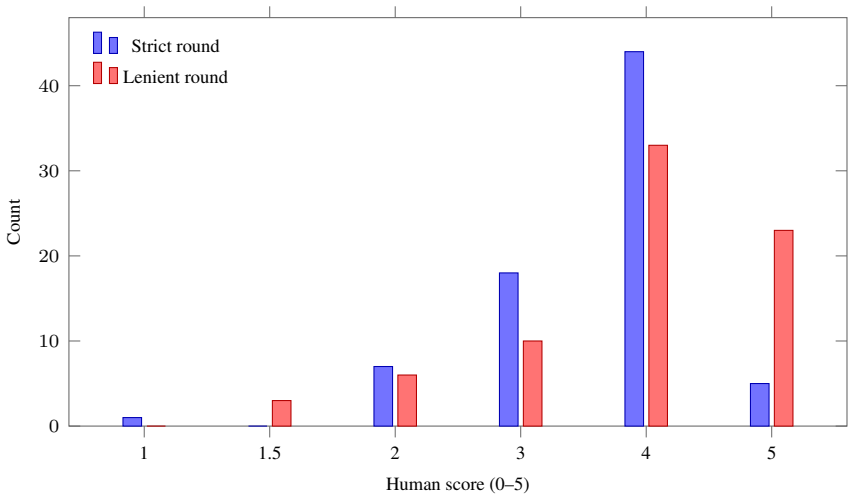


Figure 7: **Human-score dispersion across rubric rounds (75 rated pairs, 0–5 scale).** Counts are over the full 75 rated pairs (72 enter the analyses). The three 1.5 scores in the lenient round arise from duplicated legacy rows for one problem (scored 1 and 2, averaged before plotting); these duplicates are unrelated to the 3 pairs dropped from analysis, which fall out on a stale filename (Appendix A). The visible shift is consistent with the MAE/MSE statistics in Table 8.

```
0.1-0.3 weak relation
0.0 trivial/incorrect

REFERENCE IMPLEMENTATION:
{reference_impl}

GENERATED THEOREM:
{generated_theorem}

Return valid JSON only:
{"score": <float>,
 "reasoning": "<one sentence>"}
```

`BAD_STRINGY` negative control

```
You are a text similarity grader.
Compare generated theorem text to the reference implementation text.
Do NOT evaluate semantic correctness or theorem meaning.
Score ONLY based on superficial lexical/format overlap:
- high score if tokens/phrases look similar
- low score if wording/structure differs
Reference implementation:
{reference_impl}

Generated theorem:
{generated_theorem}
```

Judge setup	Scale	Main	Role
GPT-5.3-Codex prompts (P8/P4/P1)	$n{=}8$	Yes	Primary judges
GPT-5.3-Codex BAD_STRINGY	$n{=}8$	Yes	Same-model stress
Qwen3-Coder prompts	$n{=}8$	Yes	Open-weights
Qwen3-Coder BAD_STRINGY	$n{=}8$	Yes	Open-weights stress
Claude Opus 4.x + controls	$n{=}8$	Yes	Third cross-model family
GPT-4o-mini prompts	$n{=}8$	Yes	Lightweight baseline
Edit-distance + random	$n{=}8$	Yes	Non-LLM controls
Claude 3 Sonnet (narrow slice)	Small	No	Prompt diagnostics

Table 9: **Cross-model validation status (legacy).** Legacy headline bridge claims used full eight-configuration panels on the analyzed rows; exploratory Sonnet figures are withheld from headline statistics.

Threshold	Flip rate (95% CI)	Pass rate (95% CI)
0.60	0.058 [0.048, 0.069]	0.583 [0.559, 0.608]
0.70	0.049 [0.040, 0.059]	0.559 [0.537, 0.581]
0.80	0.051 [0.042, 0.061]	0.544 [0.523, 0.566]
0.85	**0.046** [**0.037**, **0.057**]	0.536 [0.515, 0.557]

Table 10: **Threshold selection summary.** Modest, consistent reduction in decision flips at stricter thresholds, with moderate coverage loss; ranking unchanged across thresholds.

```
Return ONLY a JSON object:
{"score": <float 0.0 to 1.0>,
 "reasoning": "<short reason>"}
```

I WHY JUDGE SCORES ARE SPECIFICATION-RELATIVE

A perfect judge score cannot guarantee absolute program correctness; the obstacle is computability, not benchmark design. Rice's theorem states that any non-trivial semantic property of programs in a Turing-complete language is undecidable (Rice, 1953): if a recursively enumerable gold standard captured every true semantic property of a program, it would yield a decision procedure contradicting the theorem. Gold references are therefore necessarily a curated, finite set of important specifications, and evaluations are *specification-relative*: a maximal score denotes close agreement with the audited gold formalization and rubric-conditioned human ratings, not a universal guarantee beyond that reference.

J RELATED WORK (CONTINUED)

Autoformalization. Wang et al. (2024a) present THEOREMLLAMA, enhancing LLM translation into Lean 4 via NL–FL bootstrapping and iterative proof synthesis over 100K Mathlib4-derived examples, surpassing GPT-4 by over ten points on MiniF2F-Valid/Test.

Techniques for code verification. IMPROVER (Ahuja et al., 2024) introduces a Lean-aware Chain-of-States loop with retrieval and best-of-n sampling to rewrite formal proofs with improved properties. CLOVER (Sun et al., 2024) closes the loop between code, docstrings, and formal annotations with reconstruction-based consistency checks plus SMT verification, accepting 87% of correct solutions and rejecting 100% of flawed ones on CloverBench. Zhou et al. (2025) improve general-purpose graders with privileged information (gold solutions, rubrics, annotations), reaching expert-level grading on difficult benchmarks.

Agentic frameworks and tooling. TRACE (Cheng et al., 2024) treats execution traces as gradients for generative optimization of full workflows; DSPY (Khattab et al., 2023) compiles declarative LLM pipelines with automatic prompt bootstrapping; PANTOGRAPH (Aniva et al., 2024) exposes Lean 4 proof states and tactics programmatically for learning agents.

Scalable oversight with LLM judges. Multi-agent oversight frameworks such as debate and consultancy use weaker judges to supervise stronger agents (Kenton et al., 2024; Chan et al., 2024); debate improves truth-tracking in math and code tasks. Self-critique frameworks such as SCRIT (Tang et al., 2025) bootstrap judgment skill without stronger supervisors. Validity-centered evaluation (Salaudeen et al., 2025) distinguishes measurement, claims, and evidence — construct, content, and criterion validity — and provides the reporting blueprint our construct/criterion/instrument split follows.

Robust evaluation without ground truth. Robertson & Koyejo (2025) model the overseer as an agent and propose information-theoretic, incentive-compatible queries (bounded f-divergences such as TVD-MI) with polynomial robustness to adversarial strategies. Peer-prediction consensus certifies trust by asking whether multiple judges' outputs share information about a source — a population-level protocol. Behavioral-property screening is complementary: it evaluates a single judge in isolation when consensus is unavailable (single-judge inference, novel domains, sparse panels), and a successfully bridged protocol could amortize the human-annotation cost that consensus mechanisms also incur; the present frozen bridge does not.

Consensus aggregation vs. property screening. Denisov-Blanch et al. (2026) stress-test five inference-time aggregation strategies across benchmarks and model scales; even at $25\times$ single-sample cost, no aggregator consistently beats a single sample in verifier-absent domains, and several amplify shared misconceptions rooted in structurally correlated LLM errors (in MATH plurality-wrong cases the most common incorrect answer accounts for 65–87% of wrong responses; models agree at $\kappa \approx 0.35$ even on random-ASCII controls). Their diagnosis — consensus tracks what the crowd will say, not what is true — sharpens the case for property-based, single-judge screening whenever a verifier or reference checked against known outcomes exists; our work sits on the verifier-present side of that boundary. For oracle-free extensions, TVD-MI-style cross-judge information is a robustness diagnostic, not a truth proxy.

Specializing to a test set vs. screening across configurations. The *Specialist* method (Finkelstein et al., 2025) deliberately overfits a prompted autorater to a single benchmark using same-source historical ratings as demonstrations, with large gains on WMT MQM tasks. Both lines target the expensive part of human evaluation along orthogonal axes: Specialist amortizes *within* a test set (and, as its authors show, specializes to the individual rater), while a validated property screen would amortize *across judge configurations* with no human labels on perturbed variants. Their single-rater specialization finding echoes our single-annotator ceiling; both motivate multi-rater, protocol-consistent held-out designs.

Formal methods as oversight. Zhou et al. (2024) filter autoformalized solutions via proof checking, outperforming majority voting on GSM8K; SAFE (Liu et al., 2025) verifies intermediate steps in Lean 4 and introduces FormalStep. Ospanov et al. (2023) integrate Lean feedback into repair cycles, boosting proof success on miniF2F; Miranda et al. (2026) extend the paradigm to program verification, where self-debugging agents quadruple success over one-shot prompting.

Reference-free judging of autoformalization. Zhang et al. (2025) score formalizations without a gold reference via an ensemble with a fine-grained taxonomy — scalable when golds are unavailable, but validated against subjective human preference rather than machine-checkable equivalence, and motivated by a small non-random sample of "potentially incorrect" formalizations pre-selected by an uncertified LLM. Benchmark noise rarely alters relative model rankings because errors correlate across systems (Salaudeen & Hardt, 2024), so a small number of flawed gold examples does not invalidate comparative evaluation; we proceed from kernel-checked gold references and emphasize development-panel evidence for pairwise scores.

Verified-coding benchmarks and specification generation. Bursuc et al. (2025) presume well-formed specifications and evaluate end-to-end verification under them (Dafny, Verus/Rust, Lean). Impactful problems often lack both complete specifications and ground-truth answers; VERIBENCH targets the specification-*generation* gap, and the Unsolved Questions paradigm (Nie et al., 2025) extends evaluation to specless, answerless settings via validator pipelines. Together these shift eval-

uation from "solve what is specified" to "specify, validate, then solve"; our contribution is the validation layer for the semantic judges that grading generated specifications requires.

HCI for oversight. HCI reframes oversight as appropriate reliance: interfaces that expose process and embed lightweight verification rather than one-shot answers (Bach et al., 2022; Ding et al., 2025; Wang et al., 2024b). Interaction loops that make correctness checkable during generation — test-elicitation (Naik et al., 2024), explain-then-correct (Yan et al., 2024), pair debugging (Ma et al., 2024), exposed multi-step pipelines (Wu et al., 2022), and source-grounded verification (Hennigen et al., 2024) — shift user trust from fluency to verifiable progress. Two patterns port directly to formal specs: counterfactual spec probes (remove a postcondition or inject a bug, display the score impact) and stepwise proof oversight (surface intermediate goals, mark unverified suggestions); both interface naturally with a graded, semantics-sensitive evaluator and a judge-card report in the validity-centered sense of Salaudeen et al. (2025).

Formal-verification context. With a proof assistant, oversight decomposes: the kernel deterministically checks every proof step, so the human verifies a much smaller artifact — the statement, definitions, and assumptions. Lean 4 lets runnable code, specifications, and machine-checked proofs live in one dependently-typed language with mature tooling (`mathlib4`, metaprogramming, Pantograph, Aesop), which is why VERIBENCH adopts it; its guarantees remain static and specification-relative, and theoretical limits (Rice, halting, incompleteness) bound what the verifier can settle — not the oversight claim that humans are best deployed on asking the right question. Alternatives trade differently: Dafny's SMT-centered workflow offers push-button automation but not Lean's value-indexed, higher-order specifications; TLA+ excels at protocol specification but is not executable; Verus verifies Rust with a larger trusted surface than Lean's small kernel.

Hybrid TVD-MI protocols. Where gold specifications exist, fitting mappings from TVD-MI-style information scores to known gold outcomes (compile success, unit tests, post-condition proofs) could yield composite metrics that preserve gaming-resistance while adding correctness probabilities whose predictions match observed frequencies — a practical bridge between unsupervised robustness and supervised precision (Robertson & Koyejo, 2025).

K PRE-REGISTRATION SUMMARIES

Two held-out screens were pre-registered against the revised-protocol criterion (Section 7); their documents (candidates, decision thresholds, analysis seeds, success bars, committed before new judge scores existed) ship in the supplementary archive. **Screen 1** (14 configurations): registered outcome *underpowered*. **Screen 2** (32 configurations): registered outcome *negative* — no candidate met the success bar (strongest $|\rho| = 0.39$, Holm-corrected $p = 0.25$; leave-one-out selection transfer $\rho = 0.24$, permutation $p = 0.49$). The registered next analyses are: fitting and selection under one rubric end-to-end; expansion toward ≈ 40 judge configurations; the δ-measurement collection for Route B (Appendix C); and decision-utility measurement at a fixed false-pass tolerance (Table 2).